



Intermolecular interactions governing adsorption of bulky molecules on ZIF-8: Insight from adsorption kinetics of benzene and 6-membered ring alicyclic hydrocarbons

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Abstract

This study examined the adsorption kinetics of benzene and six-membered ring alicyclic hydrocarbons on Zeolitic imidazolate framework-8 (ZIF-8) to elucidate the interactions between the adsorbed molecules and linkers on the adsorption behavior of bulky molecules. The temperature dependence of the apparent Fickian diffusion coefficient was analyzed to determine the activation entropy and energy of diffusion. The significant negative values observed for the activation entropy, from -170 to $-220 \text{ J K}^{-1} \text{ mol}^{-1}$, indicate a restriction on the degree of freedom of the adsorbate molecules within the diffusion transition state, thereby reducing the apparent diffusion coefficients. The molecular orientations of the adsorbate molecules traversing the ZIF-8 aperture were investigated using Monte Carlo simulations. Order parameter analysis revealed that the molecular orientation was constrained to minimize the energy barrier with passage through the aperture. Fourier-transform infrared (FT-IR) spectroscopy and ^{13}C cross-polarization magic angle spinning (CP/MAS) nuclear magnetic resonance (NMR) measurements supported the CH- π interactions between the adsorbate molecules and 2-methylimidazolate linkers. This interaction maintained the orientation of the adsorbate molecules, facilitating their passage through the ZIF-8 apertures. CH- π interactions influence molecular orientation control in the adsorption and diffusion of large molecules containing π electrons in porous materials with organic ligands.

Keywords Adsorption kinetics · Molecular sieving · Diffusion · Activation entropy · CH- π interaction

1 Introduction

The growing interest in hydrogen as a prospective energy source has led to increased demand for hydrogen as a renewable resource in recent years. To establish a dependable hydrogen supply, there is an urgent need to construct large-scale storage facilities, and economically viable transportation systems [1–3]. The organic hydride process is a

promising strategy for hydrogen supply [4, 5] involving the storage of hydrogen in saturated hydrocarbons such as methylcyclohexane, which can be dehydrogenated to produce unsaturated hydrocarbons like toluene for hydrogen extraction [6, 7]. The efficiency of the hydrogen extraction process can be improved by continuously removing toluene from the reaction system. Therefore, the development of molecular sieve technologies capable of effectively separating toluene from methylcyclohexane–toluene mixtures while minimizing energy consumption is essential. Beyond the methylcyclohexane/toluene system, the advancement of molecular sieve materials for the selective separation and adsorption of saturated and unsaturated compounds in bulky hydrocarbons with high hydrogen densities is anticipated to further enhance the organic hydride method [8–12].

Zeolitic Imidazolate Framework 8 (ZIF-8), a type of metal-organic framework, is one of the leading candidates for such molecular sieve materials. The ZIF-8 is represented

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by the formula $[\text{Zn}(\text{C}_4\text{H}_5\text{N}_2)_2]_n$, consisting of Zn^{2+} ions and the 2-methylimidazolate anion ($\text{C}_4\text{H}_5\text{N}_2^-$, MeIm^-) [13]. It possesses a sodalite structure characterized by micropores with a diameter of 11.4 Å, interconnected by eight 6-membered ring (6MR) openings measuring 3.4 Å in diameter [14]. This material exhibits significant potential for gas adsorption and separation applications [15, 16], which can be attributed to its exceptional molecular sieving properties stemming from its flexible framework. The rotational motion of the bridging ligand modifies the apparent size of the 6MR opening, allowing the capture of molecules larger than 3.4 Å [17, 18]. This phenomenon, referred to as “gate adsorption,” results from the framework’s flexibility [19–21]. The unique molecular sieving behavior of ZIF-8 is characterized by a nearly linear relationship between the activation energy for gas molecular diffusion and the size of the adsorbed molecules [18], although the effective diffusion coefficient derived from the adsorption rate decreases with the kinetic diameter of the adsorbed molecules. This diffusion behavior is linked to the swing effect, which is induced by the torsional motion of the imidazole ligand around the Zn–Im–Zn axis [22–24]. Significant fluctuations in the bridging ligands have been confirmed using various spectroscopic techniques, including far-infrared absorption spectroscopy [25], inelastic neutron scattering [26], terahertz spectroscopy [27], and solid-state NMR [28, 29] methods.

However, bulky molecules that exhibit significant molecular shape anisotropy may adopt an energetically favorable orientation for navigating through the 6MR aperture in an orientation that minimizes the molecular cross-sectional area. In previous studies, it was observed that for benzene and its derivatives (toluene and xylene), the minimum molecular cross-section rather than the average molecular diameter predominantly influenced the molecular orientation of the adsorbed molecules as they traversed the 6MR aperture [30, 31]. Recently, Poryvaev et al. reported that benzene was adsorbed onto ZIF-8 with an adsorption selectivity of approximately 270 from a benzene/cyclohexane solution containing >90% cyclohexane [32]. Their findings indicated that the interaction between benzene and cyclohexane with the 6MR aperture of ZIF-8 was governed by the molecular thickness of the adsorbed molecules which is corresponding to the molecular dimension “MIN-1” proposed by Webster et al. [43] (see Supplementary Information). This fact suggested that the molecular orientation was critical for the passage of benzene and cyclohexane through the 6MR aperture. Thus, for adsorbed molecules with bulky and highly anisotropic geometries, both the intermolecular interactions with the 6MR aperture (enthalpic contributions) and the molecular orientation during passage through the

aperture (entropic contributions) were expected to significantly influence their diffusion behavior.

To sustain a favorable molecular orientation for passage through the 6MR aperture, highly directional interactions likely govern the orientation of the adsorbed molecules, as opposed to isotropic intermolecular interactions such as dispersion forces. Identifying the intermolecular interactions that contribute to the maintenance of molecular orientation is anticipated to enhance the theoretical understanding of the adsorption behavior of bulky molecules on ZIF-8, particularly regarding diffusion phenomena. Potential candidates for such interactions include hydrogen bonds, tetrel bonds, CH- π interactions, and halogen- π interactions [33, 34]. We hypothesized that the double bond of 2-methylimidazolate, the bridging ligand of ZIF-8, would facilitate CH- π interactions.

This study aimed to elucidate the molecular-level adsorption mechanism of bulky hydrocarbons in ZIF-8. Initially, we focused on the adsorption kinetics of benzene and several 6-membered ring alicyclic hydrocarbons on ZIF-8, with an emphasis on evaluating the influence of intermolecular interactions on diffusion behavior. The selected alicyclic compounds (1,3-cyclohexadiene, 1,4-cyclohexadiene, cyclohexene, and cyclohexane, summarized in Scheme S1 in supporting information) had a similar projected molecular diameter to benzene but differed in molecular thickness and π -electron content (see Table S1, in SI). These variations facilitate the analysis of the diffusion coefficients and activation parameters in relation to the molecular dimensions and the presence of π -electrons. This study employed Monte Carlo simulations to investigate the critical process of molecules traversing the 6MR aperture of ZIF-8, with particular emphasis on the orientation of bulky molecules during this process. Additionally, Fourier-transform infrared absorption spectroscopy and ^{13}C cross-polarization magic-angle sample spinning (CPMAS) NMR techniques were utilized to assess adsorbate-adsorbent interactions, aiming to identify specific intermolecular interactions that influence molecular orientation as the adsorbate molecules navigate through the 6MR aperture. These research strategies collectively offer molecular-level understanding of the adsorption mechanism of bulky hydrocarbon molecules on ZIF-8.

2 Experimental

2.1 Chemicals

The ZIF-8 (Basolite Z1200, BASF) and 1,3-cyclohexadiene (97% purity) were obtained from Sigma-Aldrich. Benzene (purity >99.5%), cyclohexane (purity >99.5%), 1,4-cyclohexadiene (purity >95.0%), and cyclohexene

(purity > 97.0%) were purchased from Fujifilm Wako Pure Chemicals Co., Ltd. The reagents utilized were Wako Special Grade for benzene and cyclohexane and Wako 1st Grade for 1,4-cyclohexadiene and cyclohexene. All the reagents were used without further purification. For this study, benzene, 1,3-cyclohexadiene, 1,4-cyclohexadiene, cyclohexene, and cyclohexane are referred to as BZ, 1,3-CHD, 1,4-CHD, CHEN, and CHEX, respectively.

2.2 Characterization

2.2.1 Nitrogen adsorption isotherm

Prior to use, ZIF-8 was activated by heating under vacuum (0.1 Torr) at 150 °C for 4 h. Nitrogen adsorption isotherms were recorded using an Autosorb-iQ apparatus (Anton Paar GmbH) to determine the pore volume and BET surface area of ZIF-8. The adsorption and desorption branches were recorded over a relative pressure (p/p_0) range of 0.000006–0.987, where p_0 represents the saturated vapor pressure of nitrogen at its boiling point of liquid nitrogen (77 K). The porosity of ZIF-8 was evaluated from the nitrogen adsorption isotherm (Figure S1, Supplementary Information), which exhibited a characteristic Type I (a) pattern according to the IUPAC classification. The BET surface area was calculated to be 1590 m² g⁻¹. The pore volume of 0.62 cm³ g⁻¹ was evaluated from the adsorption amount of N₂ at 0.96 of p/p_0 .

2.2.2 Scanning electron microscopy (SEM)

Scanning electron microscopy (SEM) was performed using a field-emission scanning electron microscope SU6600 (Hitachi High-Technologies Co.) to analyze the size distribution and morphology of the crystallites. The SEM images are shown in Figure S2 (Supplementary Information). To analyze the particle distribution in ZIF-8, SEM images at a magnification of 10,000 times were taken at 14 distinct points on the sample, resulting in a statistical size distribution of 240 particles. The average particle diameter, based on the crystalline size distribution (Figure S3), was determined to be 450 nm, corresponding to a radius of 225 nm.

2.2.3 Powder X-ray diffraction (PXRD)

Powder X-ray diffraction (PXRD) patterns were measured using a MiniFlex600 powder X-ray diffractometer (Rigaku Co., Ltd.) in the range of 2θ from 4° to 40° with a step of 0.005° at room temperature. Diffraction was conducted using Cu K α irradiation with a wavelength of 1.5418 Å. The PXRD patterns are presented in Figure S5 (see SI).

2.2.4 Fourier transferred infrared (FT-IR) absorption spectroscopy

Fourier-transform infrared (FT-IR) measurements were performed using an FT/IR-4600AC spectrometer (JASCO Co.) over the wavenumber range of 400–4000 cm⁻¹ at room temperature with a resolution of 1 cm⁻¹. To prevent the amorphization of the ZIF-8 sample due to uneven pressure during mortar grinding, KBr pellet samples were prepared by compressing the powder sample between two KBr plates. Spectra were recorded in transmittance units and subsequently converted to absorbance values.

2.2.5 Solid-state ¹³C NMR spectroscopy

High-resolution solid-state ¹³C NMR spectra were obtained using a Bruker Avance III 400WB spectrometer equipped with a superconducting magnet (9.4 T), which provided resonance frequencies of 400.25 and 100.64 MHz for the ¹H and ¹³C nuclei, respectively. Free induction decay (FID) signals for ¹³C nuclei were acquired using a single-pulse sequence with ¹H decoupling under magic-angle sample spinning (MAS) conditions and a ¹H decoupling MAS cross-polarization (CP) pulse sequence. The powdered samples were packed in a 4 mm-diameter cylindrical zirconia rotor and spun at 10 kHz for MAS. Adamantane was used as an external standard for the ¹³C chemical shift, with a CH₂ peak referenced at 38.5 ppm (relative to tetramethylsilane).

2.3 Measurement of time dependence of adsorption uptake

The time dependence of the adsorption uptake was measured according to the procedure reported in our previous study [29–31]. The procedure is outlined as follows.

The ZIF-8 powder sample underwent pretreatment (heating at 150 °C in a vacuum of 0.1 Torr for 12 h). The samples with an average weight of 7–15 mg were placed in an aluminum pan (5 mm × 3 mm depth). The aluminum pan was placed in a hermetically sealed vessel containing 3 mL of the appropriate solvent. The presence of air in the container had a minimal effect on solvent vapor adsorption, as nitrogen (N₂) and oxygen (O₂) behave as supercritical gases at ambient temperatures. The adsorbates used in this study were BZ, 1,3-CHD, 1,4-CHD, CHEN, and CHEX. The vessel was affixed to a base constructed from thermally insulating styrofoam and placed within a temperature-controlled enclosure with an accuracy of ±0.5 °C. The time dependence of adsorption uptake for each adsorbate was conducted at 30, 40, and 50 °C. The vapor pressure of the applied adsorbate at each temperature is shown in Figure S4 (Supplementary Information). At a constant temperature, BZ, 1,3-CHD,

and CHEX exhibited similar vapor pressures within ± 1 kPa, whereas CHEN and 1,4-CHD displayed slightly lower vapor pressures, with differences of 1.5 kPa for CHEN and 5 kPa for 1,4-CHD. The samples were weighed at intervals until a stable value was reached. The increase in the sample weight was measured simultaneously for the three samples, and their average was considered as the measured value. The experimental errors were estimated based on the deviation from the average, exhibiting an accuracy of $\pm 5\%$.

2.4 Monte carlo simulations

A Monte Carlo simulation was performed using proprietary software (rxmc) developed at Shinshu University. The simulation was conducted at a constant temperature of 298 K under NVT conditions with a minimum of 100,000 trials. In the MC simulation, the maximum translational displacement was 0.80 Å and the maximum rotational displacement was 0.40 rad. No reinsertion attempts were performed.

The force field parameters, which include Lennard-Jones parameters and partial charges assigned to the atoms forming the 6MR aperture, were utilized in the molecular dynamics (MD) simulations of ZIF-8 [35, 36]. In this study, the non-bonded force field parameters for the flexible framework of ZIF-8, as proposed by Zheng et al. [36], were employed, whereas the OPLS-AA force field parameters were applied to the adsorbed molecules [38, 39]. The adsorbed molecules were treated as rigid molecules, and an all-atom model was used. The parameters used are listed in Tables S3 and S4 in the SI. The GCMC simulation with these force fields successfully reproduced the benzene experimental desorption isotherm in both the Type I isotherm shape and saturated adsorption amount.

The configuration of the adsorbed molecule was determined by optimizing its structure using density functional theory (DFT) calculations. The Gaussian 16 software package [37] was employed, utilizing the B3LYP functional and 6-31G(d) basis set for the calculations. The 6MR structure employed in the simulation was modeled by extracting one-eighth of the unit cells dimension ($a/2$, $a/2$, $a/2$) from the ZIF-8 crystal (lattice constant a). The vector from the lattice point (0, 0, 0) to (1/2, 1/2, 1/2) was designated as the Z-axis in the reference coordinate system. The origin of this reference coordinate system was set at the center of the 6MR plane (lattice point (1/4, 1/4, 1/4)). The direction toward lattice point (0, 0, 0) was defined as negative (−), and that toward (1/2, 1/2, 1/2) as positive (+). The 6MR structure is shown in Figure S6. At a specified Z value, the mass center of the adsorbed molecule was displaced over a $4 \text{ Å} \times 4 \text{ Å}$ plane centered on the reference coordinate axis. The most energetically favorable configuration was then determined using Monte Carlo simulation. The opening of the 6MR aperture was adjusted by systematically rotating the tilt angle of the 2-methylimidazole bridging ligand along its N-N axis. The tilt angle was defined as the angle between the molecular plane of the bridging ligand and the 6MR aperture plane. Variations in tilt were expressed as positive ($+\Delta\theta$) or negative ($-\Delta\theta$) deviations from the ligand's original orientation of the crystal lattice. The adsorbed molecule was moved along the Z-axis in the reference coordinate system from $+4 \text{ Å}$ to -4 Å in 0.2 Å increments, and the potential energy between the adsorbed molecule and the 6MR for each Z value was evaluated by varying $\Delta\theta$. For the order parameter analysis, the simulations were carried out in the Z value from $+5 \text{ Å}$ to -5 Å .

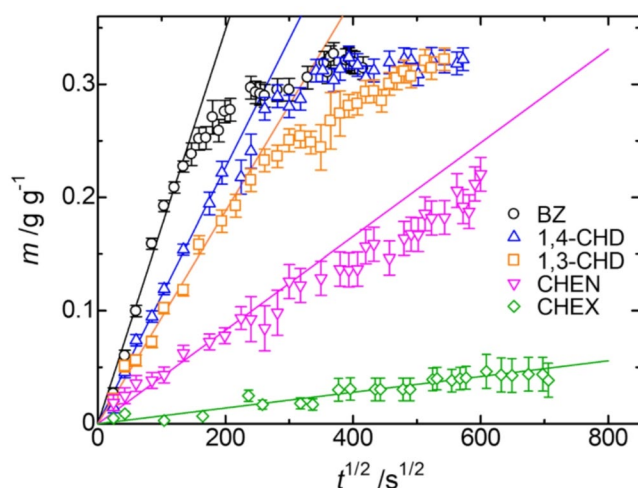


Fig. 1 Time dependence of the relative mass of benzene and six-membered ring alicyclic hydrocarbons adsorbed on ZIF-8 at 303 K. The solid lines represent the results of fitting Eq. (1) to the linear section of each dataset

3 Results and discussion

3.1 Time dependence of adsorption uptake

Figure 1 depicts the time-dependence of the adsorption uptake of individual hydrocarbons on ZIF-8 at 303 K, illustrating the relationship between the amount of adsorption and the square root of the adsorption time. Initially, all the vapor species exhibited a linear increase in the adsorption amount, which was proportional to the square root of time, suggesting that micropore diffusion was the primary mechanism governing the adsorption of individual solvent vapors on ZIF-8. These observations can be characterized using the Fickian diffusion model, which is applicable to spherical particles [40]. Assuming that the concentration of adsorbed molecules at the center of the particle is negligible (i.e., in the short diffusion time region), the model can be approximated as

$$m(t) = m_{\text{ext}} + m_{\infty} \frac{6}{\sqrt{\pi}} \sqrt{\frac{D_{f,\text{app}} t}{r_{\text{ave}}^2}} \quad (1)$$

where $D_{f,\text{app}}$ denotes the apparent Fickian diffusion coefficient, which is derived from the initial slope of the time-dependent adsorption curve; r_{ave} represents the average radius of the spherical particles; m_{∞} indicates the saturation adsorption amount; and m_{ext} refers to the adsorption amount present on the surface of the particle [41]. In general, the diffusion coefficient is dependent on the equilibrium pressure. The true Fickian diffusion coefficient D_{f0} is the value extrapolated to zero vapor pressure. Thus, under finite vapor pressure, the diffusion coefficient D_f is represented as $D_{f0}/(1-\theta)$, where $(1-\theta)$ is a correction factor, and θ is the pore filling ratio. However, in this study, all uptake measurements were conducted near the saturated vapor pressure at each temperature, suggesting that the correction factors for each adsorbate were nearly equivalent. Therefore, the obtained diffusion coefficients reflect the relative mass transfer of each adsorbate into ZIF-8. In this study, consequently, we represent them as apparent Fickian diffusion coefficients $D_{f,\text{app}}$ and discuss their relative relationships with each other.

The diffusion coefficient $D_{f,\text{app}}$ can be calculated from the slope of the linear portion depicted in Fig. 1, assuming that the saturation adsorption amount is known. The data indicated that BZ and 1,4-CHD achieved saturation during adsorption, whereas 1,3-CHD approached saturation. In contrast, the CHEN and CHEX did not reach saturation during this period. For convenience, the adsorption quantity recorded at the longest adsorption time was designated as the saturation adsorption quantity, m_{∞} , at equilibrium. The $D_{f,\text{app}}$ values for each adsorbed molecule were extracted from the linear segments, as shown in Fig. 1. Figure S7 (SI) presents the time-dependent adsorption amounts at 313 and 323 K. Table 1 summarizes the $D_{f,\text{app}}$ values obtained at 303, 313, and 323 K. Each adsorbate exhibited the different $D_{f,\text{app}}$ value. Notably, it was interesting that despite being structural isomers with identical molecular weights, 1,3-CHD and 1,4-CHD showed different uptake rate, providing distinct $D_{f,\text{app}}$ value.

Table 1 Diffusion coefficients of adsorbate molecule determined from the initial slope of adsorption uptake curve

Adsorbate	$D_{f,\text{app}} / \text{m}^2 \text{s}^{-1}$		
	303 K	313 K	323 K
BZ	$1.3 (1) \times 10^{-19}$	$2.2 (1) \times 10^{-19}$	$2.8 (1) \times 10^{-19}$
1,4-CHD	$5.6 (1) \times 10^{-20}$	$7.9 (1) \times 10^{-20}$	$9.5 (1) \times 10^{-20}$
1,3-CHD	$3.2 (1) \times 10^{-20}$	$3.5 (1) \times 10^{-20}$	$5.4 (1) \times 10^{-20}$
CHEN	$1.2 (1) \times 10^{-20}$	$1.8 (1) \times 10^{-20}$	$2.2 (1) \times 10^{-20}$
CHEX (chair)	$8.2 (1) \times 10^{-21}$	$1.1 (1) \times 10^{-20}$	$1.7 (1) \times 10^{-20}$

The member in parentheses presents the experimental error

When larger molecules are adsorbed onto ZIF-8, they must navigate through the 6MR aperture, which consists of six zinc (Zn) atoms and 2-methylimidazole bridging ligands. A previous study established that, for benzene and its derivatives adsorbed on ZIF-8 [30], traversal through the 6MR aperture represents the rate-limiting step in the diffusion process. Furthermore, the movement of bulky molecules through the 6MR aperture is primarily governed by configurational diffusion, which is characterized by a restricted molecular orientation and rotation. In addition, The PXRD pattern for each sample indicated that the crystal lattice of ZIF-8 retained its structure even after the confinement of each adsorbate within the micropores of ZIF-8 (see Figure S5). However, there was an observed increase in reflections from the (211), (310), and (222) planes in the crystal lattice with space group $I \bar{4} 3m$. The (211) and (310) planes intersect the vacancy of the ZIF-8 unit cell, while the (222) plane traverses the entrance of the 6-membered ring aperture, where the micropores are interconnected. This suggests that the vacancies in ZIF-8 are filled with the adsorbed molecules without causing a collapse of the ZIF-8 crystal lattice. Consequently, the temperature dependence of $D_{f,\text{app}}$ should be assessed using the following equation derived from Eyring's transition state theory [42] rather than the traditional Arrhenius equation:

$$D_{f,\text{app}} = e\lambda^2 \frac{kT}{h} e^{\Delta S^\ddagger/R} e^{-E_a/RT} \quad (2)$$

where E_a signifies the diffusion activation energy, $\Delta S^\ddagger$ represents the diffusion activation entropy, h denotes Planck's constant, k is Boltzmann's constant, and λ indicates the distance between the molecular jumps. During the diffusion process, molecules jump between adjacent lattices via 6MR apertures with $\lambda = 3^{1/2}a/2$. The lattice constant a is 1.701 nm at 298 K [22], resulting in a λ of 1.473 nm.

Figure 2 illustrates a logarithmic representation of $D_{f,\text{app}} \cdot T^{-1}$ versus the reciprocal temperature (Eyring plot). The slope and intercept of the graph correspond to the activation energy and entropy, respectively. The activation parameters for each adsorbed molecule are listed in Table 2.

The correction term, $1-\theta$, contributes $R\ln(1-\theta)$ to the activation entropy obtained from $D_{f,\text{app}}$. This contribution depends on the value of $(1-\theta)$, extending to tens of $\text{J K}^{-1} \text{mol}^{-1}$. Therefore, the value of $\Delta S^\ddagger$ determined from the Eyring plot should be reported as two significant figures.

Figure 3 presents the molecular size dependence of $D_{f,\text{app}}$ at 303 K and the activation parameters E_a and $-\Delta S^\ddagger$ for individual hydrocarbons. The semi-logarithmic plot of $D_{f,\text{app}}$ shows a linear decrease with increasing molecular size, whereas the standard plots of E_a and $-\Delta S^\ddagger$ exhibited a V-shaped dependence on the molecular size. Specifically,

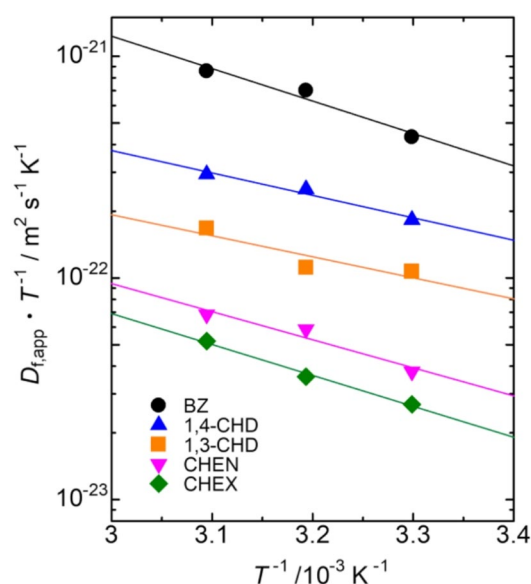


Fig. 2 Eyring plot of the diffusion coefficients of all hydrocarbon molecules

Table 2 Average molecular size and activation parameters determined from the temperature dependence of the diffusion coefficients

Adsorbate	$\sigma_{\text{average}}/\text{nm}$	$E_a/\text{kJ mol}^{-1}$	$\Delta S^\ddagger/\text{J K}^{-1} \text{mol}^{-1}$
BZ	0.525	28 (2)	-170 (10)
1,4-CHD	0.537	19 (2)	-210 (10)
1,3-CHD	0.538	18 (2)	-220 (10)
CHEN	0.548	24 (2)	-210 (10)
CHEX (chair)	0.558	27 (1)	-200 (10)

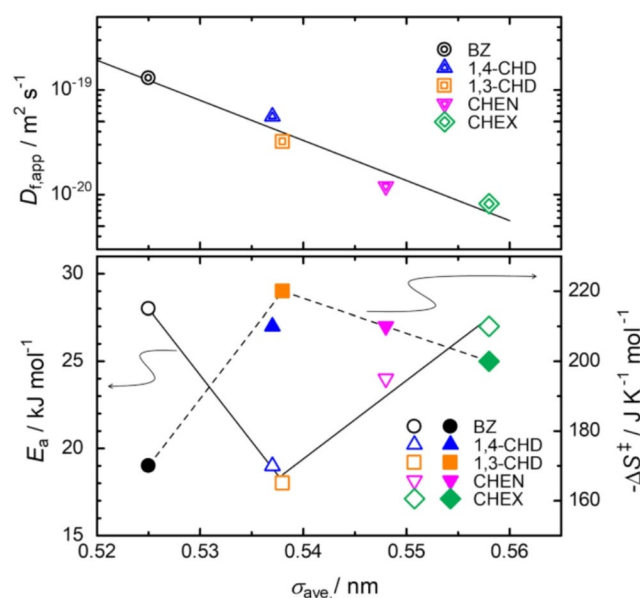


Fig. 3 Molecular size dependence of Fickian diffusion coefficients, $D_{f, \text{app}}$, for adsorbate molecules at 303 K and the activation parameters, E_a (open symbols) and $-\Delta S^\ddagger$ (filled symbols)

E_a reached its minimum for 1,4-CHD and 1,3-CHD, which possessed intermediate molecular sizes, and attained its maximum values for BZ and CHEX, representing the smallest and largest molecular sizes, respectively. This observation suggests that the passage of adsorbed molecules through 6MR aperture is governed by the combination of molecular size and specific host-guest interactions at the 6MR aperture site. The activation energy E_a can be lowered by stabilizing the transition state or raised by stabilizing the adsorbed molecules within the pores. A lower E_a enhances the likelihood of adsorbed molecules overcoming the diffusion energy barrier at a specific temperature. Conversely, $\Delta S^\ddagger$ exerts an opposing influence on $D_{f, \text{app}}$. The substantial negative activation entropy indicates a pronounced restriction on the degrees of freedom of the adsorbed molecule within the diffusion transition state [31], which leads to a reduced diffusion frequency. In BZ, the effect of $\Delta S^\ddagger$ —despite having the smallest absolute value—surpasses that of the largest E_a , resulting in the highest $D_{f, \text{app}}$ value among the compounds. For 1,4-CHD and 1,3-CHD, the E_a values are less than 20 kJ mol^{-1} ; however, their $-\Delta S^\ddagger$ values reach up to ca. $210 \text{ J K}^{-1} \text{mol}^{-1}$. The competing effects of these opposing factors may result in the contribution of activation entropy surpassing that of activation energy, thereby causing a decline in $D_{f, \text{app}}$ as the molecular size increases. In BZ, 1,4-CHD, and 1,3-CHD, E_a and $\Delta S^\ddagger$ exhibited a trade-off relationship. In contrast, for CHEN and CHEX, both the high E_a values and the relatively large absolute values of $\Delta S^\ddagger$ decrease the $D_{f, \text{app}}$ values. In the following sections, we examine the pronounced effects of molecular orientation on the diffusion of adsorbed molecules through the 6MR and identify the specific intermolecular interactions that significantly influence the E_a .

3.2 Monte Carlo simulation for 6MR passage of adsorbate molecules

Monte Carlo simulations were employed to elucidate the potential energy profile and molecular orientation of the adsorbed molecules as they traverse the 6MR aperture. In this context, the 6MR aperture, characterized by a tilted angle ($\Delta\theta=0$) as derived from the crystal structure, presents an activation energy required for molecular passage with activation energies ranging from several hundred to several thousand kJ mol^{-1} . Consequently, it is nearly impossible for molecules to pass through the aperture using only the thermal energy available at room temperature. This observation implies that the 6MR aperture must undergo enlargement to facilitate the passage of target molecules. To quantify this, we evaluated the activation energy required for molecular passage as a function of the tilted angle of the bridging ligand by systematically varying this angle. By employing

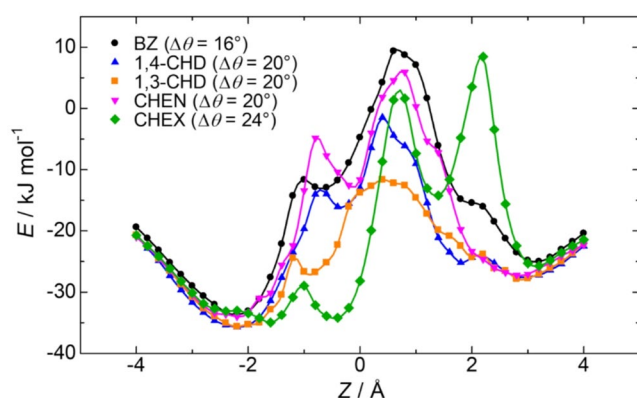


Fig. 4 Potential energy profiles obtained through Monte Carlo simulation for the adsorbed molecules as they pass through the 6MR aperture

Table 3 Experimental and simulated activation energies of adsorbed molecules passing through the 6MR aperture, tilted angles of the bridging ligand, and average diameters of the 6MR aperture

Adsorbate	ΔE / kJ mol ⁻¹	$\Delta\theta_c$ / deg.	$d_{p,ave}$ / nm	MIN-1 ^b /nm	MIN-2 ^b /nm
BZ	27.6	17.0	0.47	0.3277	0.6628
1,4-CHD	19.3	21.1	0.49	0.3845	0.6612
1,3-CHD	17.5	19.9	0.49	0.3845	0.6612
CHEN	24.2	20.55	0.49	0.4414	0.6595
CHEX (chair)	27.0 ^a	24.05	0.52	0.4982	0.6580

^a This value corresponds to the apparent activation energy when adsorption pre-equilibration occurs: $\Delta E = E_{a,a} - E_{a,d} + E_{a,in}$

^b Critical dimensions of adsorbate molecule for entry into zeolite pores. In the slit-shaped pores, the size of the adsorption in the minimum dimension, MIN-1, was determined. In the cylindrical pores, the size of the molecule in two directions is considered: the minimum dimension, MIN-1, and the next smallest dimension, MIN-2

polynomial interpolation on the acquired data, we identified a tilted angle that aligned with the experimentally measured values. Figure 4 illustrates the energy profile associated with passage through the 6MR aperture, corresponding to the experimental activation energy for each adsorbate, as determined by the time dependence of the adsorption uptake. Two minima were observed on either side of the 6MR aperture. The energy minimum at the CH₃ neck ($Z/\text{\AA} \sim -2$) was deeper than that at the CH neck ($Z/\text{\AA} \sim +3$). This finding suggests that the passage of molecules trapped in the CH₃ neck through the 6MR aperture is energetically unfavorable. Consequently, the passage of molecules through the 6MR aperture is only considered when they approach from the CH neck side [30]. In this case, the energy barrier encountered by an adsorbed molecule navigating through the 6MR aperture is defined as the difference ($\Delta E = E_{CH, \max} - E_{CH, \min}$) between the maximum energy ($E_{CH, \max}$) located near the CH neck ($+0.5 \leq Z/\text{\AA} \leq +1$) and the minimum energy ($E_{CH, \min}$) at the CH neck ($Z/\text{\AA} \sim +3$). Notably, in the case of CHEX, an additional energy maximum was detected on the

outside of the CH neck ($Z/\text{\AA} \sim +2$). The presence of the dual maxima in the energy profile aligns with the findings from previous studies. This behavior supports the use of a model that incorporates pre-equilibrium adsorption at the CH neck, represented as: $A(g) + M(s) \rightleftharpoons AM^*(s) \rightleftharpoons AM(s)$ [30]. In this model, the apparent activation energy can be expressed as $\Delta E = E_{a,a} - E_{a,d} + E_{a,in}$, where $E_{a,a}$ denotes the energy barrier for pre-adsorption at the 6MR aperture, $E_{a,d}$ is the energy barrier for the desorption of the pre-adsorbed molecule, and $E_{a,in}$ is the energy barrier for passage through the CH neck. The activation energy evaluated by MC simulation, the resulting tilted angle of the bridging ligand, and the aperture opening diameter are listed in Table 3. The correlation between the tilted angle and the opening diameter of the 6MR aperture, which reproduces the experimental values, appears to align more closely with MIN-1 than with MIN-2, as proposed by Webster et al. [43] (see Scheme S1 and Table S1 in the SI). This suggests that the energy barrier for the passage of the adsorbed molecule through the 6MR aperture is significantly influenced by the cross-sectional area of the adsorbed molecule, indicating that the molecular orientation is highly constrained during this process. This conclusion was consistent with the findings of our previous study.

To further analyze the orientation of the adsorbed molecules, we defined two order parameters:

$$S_{\perp} = \frac{1}{2} (3\langle \cos^2 \alpha \rangle - 1) \quad (3)$$

$$S_{\parallel} = \frac{1}{2} (3\langle \cos^2 \beta \rangle - 1) \quad (4)$$

where α and β defined the angles formed by the vectors perpendicular and parallel to the molecular plane, respectively, with respect to the perpendiculars passing through the center of the 6MR aperture. The characterization of these vectors and their orientation within the reference coordinate system are shown in Figure S8 (see SI). Figure 5 presents the order parameter for each molecule as it traverses the 6MR aperture. An animation depicting the molecular passage through the 6MR is provided in the SI. When the molecular orientation is isotropically distributed, the order parameter is zero. However, as each molecule passed through the aperture, both $S_{\perp}$ and $S_{\parallel}$ deviated from zero, indicating that the orientation of the molecules was constrained by the aperture. Notably, the negative value of $S_{\perp}$ suggests that the molecules navigate the aperture in an orientation in which the molecular plane is nearly perpendicular to the aperture plane. This behavior was exhibited by all the adsorbate molecules. This indicated that the passage of adsorbed molecules through the 6MR aperture was accompanied by significant orientational constraints. In other words, the translational and

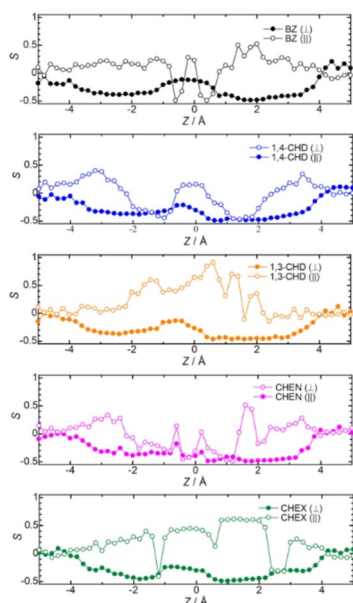


Fig. 5 Order parameters for adsorbate molecules while passing through the 6MR aperture

rotational degrees of freedom of the adsorbed molecules were considerably restricted during their traversal through the 6MR aperture. Consequently, the observed significant negative activation entropy can be attributed to the transition of molecules in the vapor to a diffusion transition state, in which the molecular orientation is markedly constrained at the 6MR aperture.

Conversely, $S_{\parallel}$ represents the variation in the rotational orientation of the molecules around an axis perpendicular to the molecular plane and exhibited different behaviors for each adsorbate molecule in the range $-3 \leq Z/\text{\AA} \leq +3$. For BZ, 1,4-CHD, CHEN, and CHEX, the values ranged from $-0.5 < S_{\parallel} < +0.5$, whereas for 1,3-CHD, $0 < S_{\parallel} < +1$. In BZ and 1,4-CHD, $S_{\parallel}$ exhibited a bipolar change centered at $Z=0$, indicating that the molecule turned its orientation around an axis perpendicular to the molecular plane and then turned it back. In the CHEN and CHEX, the variation in $S_{\parallel}$ showed a similar trend, although the range of the Z -axis was narrow. This feature implies that these molecules undergo twisting and wiggling around the axis perpendicular to the molecular plane when traversing the 6MR aperture (see the animation provided in the SI). In contrast, for 1,3-CHD, $S_{\parallel}$ varied continuously from 0 to +1 during the passage through 6MR, suggesting that the molecular orientation around the perpendicular axis changes progressively throughout the diffusion process. That is, the 1,3-CHD molecule rotates around the axis as it moves through the aperture (see the animation in the SI). This rotation is believed to minimize steric hindrance within the 6MR aperture by reducing friction between the molecule and the framework. This behavior differs from that of other molecules and may partly explain

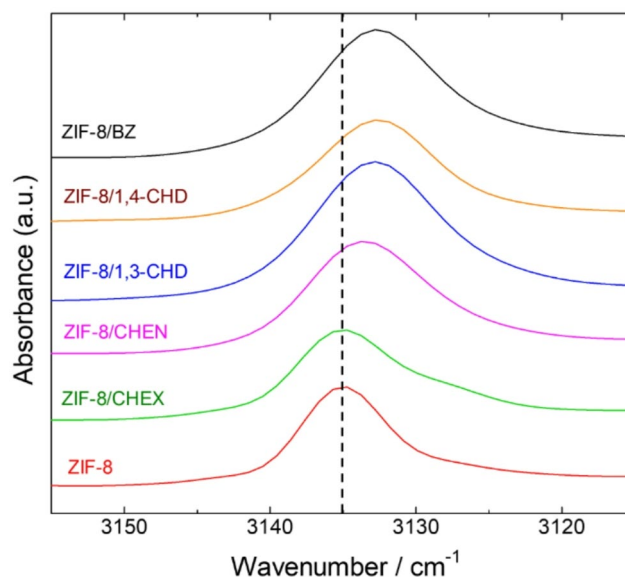


Fig. 6 FT-IR spectra of the symmetric C–H stretching band of the 2-methylimidazole group in ZIF-8 without guests and after saturation with each hydrocarbon

why 1,3-CHD exhibits a lower activation energy for diffusion than the other molecules. Distinguishing between 1,3-CHD and 1,4-CHD is crucial for understanding the factors influencing the uptake rate into ZIF-8. The differences in the E_a and $S_{\parallel}$ behaviors observed in the MC simulations between 1,3-CHD and 1,4-CHD may reflect the slight differences in the molecular cross-sections and arrangement of π bonds within the molecules.

3.3 Analysis of adsorbate-adsorbent interactions by spectroscopies

3.3.1 FT-IR spectroscopy

Figure 6 displays the symmetric C–H stretching bands associated with the 2-methylimidazole group. The full wavenumber regions of the infrared spectra are shown in Figure S9 (SI). Notably, no shift was observed in the band upon cyclohexane adsorption, while a pronounced redshift was observed over the wavenumber range of $1\text{--}2\text{ cm}^{-1}$ upon benzene and cyclic alkene adsorption. This shift was more pronounced for 1,3-CHD, 1,4-CHD, and BZ than for CHEN. This observation suggests the elongation of C–H bond, which is attributed to the attractive interaction between the π -electrons of the adsorbed molecules and the C–H bond of 2-methylimidazole [31]. One plausible attractive interaction is the CH- π interaction, which has been reported for various organic compounds, metal complexes, and molecular assemblies [33]. It has been established that CH- π bonding induces stretching of the C–H bonds, resulting in a redshift of the CH stretching band [44, 45]. Additionally, the protons

of the methyl group may function as proton donors in CH- π interactions, thereby leading to an anticipated redshift of the CH symmetric stretching band in the methyl group of the 2-methylimidazolate moiety [31]. However, the symmetric CH stretching band of the methyl group overlaps with the CH stretching bands of the adsorbate molecules. Deconvolution of the infrared bands potentially confirmed the redshift of the CH symmetric stretching band of the methyl group. However, the accuracy of the resultant redshift from this deconvolution was insufficient for a meaningful discussion of the minor wavenumber shifts observed in this study. The elongation of the C–H bonds and the corresponding redshift of the CH symmetric stretching band were also qualitatively corroborated by the molecular assembly of 2-methylimidazole with benzene, 1,3-CHD, and 1,4-CHD, optimized using DFT calculations (Figure S10, Figure S11, and Table S5). Generally, intermolecular CH- π interactions occur within 3 Å [33] and exhibit a high degree of directionality, with the orientation of the C–H bond aligned with that of the π -orbital. Consequently, the C–H and π bonds tend to be oriented nearly perpendicular to each other. Following the formation of the CH- π bond, the adsorbed molecules were arranged such that their molecular planes were positioned at an angle that was nearly perpendicular to the 6MR aperture plane. This orientation facilitates the ingress of adsorbed molecules into the 6MR apertures. Thus, the CH- π interaction between the 2-methylimidazole ring and the adsorbed molecules is crucial for governing the molecular orientation necessary for the effective passage through the 6MR aperture for benzene and cyclic alkenes.

3.3.2 ^{13}C CP/MAS NMR spectroscopy

The ^{13}C CP/MAS NMR spectra of ZIF-8, both in its pure form and with various adsorbates are shown in Fig. 7. Cross-polarization (CP) measurements are particularly effective for highlighting immobile molecular segments within solid matrices while attenuating signals from more mobile components. Consequently, the highly mobile adsorbed molecules appeared as minor signals in the spectra. The predominant peaks are attributed to the 2-methylimidazole moiety, suggesting that the ZIF-8 framework exhibited greater rigidity and reduced mobility than the adsorbate molecules. Specifically, the resonance lines at approximately 14, 124, and 151 ppm correspond to the methyl group, CH carbon of the imidazole ring, and quaternary carbon of the 2-methylimidazolate moiety, respectively. Figures 8(a), (b), and (c) provide enlarged views of these spectral components. Notably, the adsorption of benzene and cyclic alkenes resulted in down-field shifts of 0.5–1.3 ppm in the resonance lines associated with the methyl group and CH carbon of the imidazole ring, whereas the adsorption of cyclohexane did not elicit such

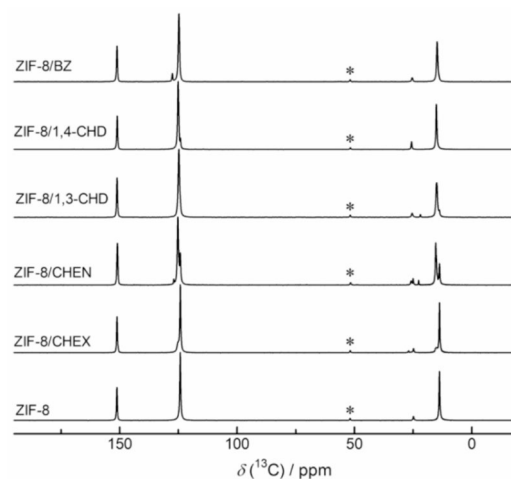


Fig. 7 ^{13}C CP/MAS NMR spectra of ZIF-8 and ZIF-8 with various adsorbate molecules. The asterisk indicates the spinning sideband. The minor peaks correspond to the adsorbate molecules

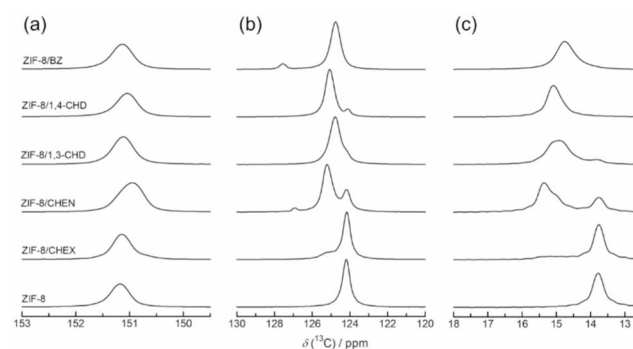


Fig. 8 Extended view of ^{13}C CPMAS NMR spectra in the chemical shift region of the imidazolate quaternary carbon (a), imidazolate ring CH carbon (b), and methyl carbon (c). The adsorption of benzene and cyclic alkenes induced low-field shifts of 0.5–1.3 ppm in the resonance lines of the methyl group and CH carbon of the imidazole ring

shifts. The resonance lines corresponding to quaternary carbons remained largely unchanged after adsorption. These observations suggest a significant interaction between the π -electrons of the adsorbed molecules and both the methyl group and the CH carbon of the 2-methylimidazolate moiety. Our previous studies indicated that both the methyl and CH protons of the imidazole ring can act as proton donors in CH- π bond. According to Pople's point magnetic dipole model [46, 47], the shielding effect of carbon along a perpendicular line through the center of the molecular plane of the benzene ring is described by the equation $\Delta\sigma_{\text{C}} = 51.96/R^3$, where R represents the average distance between the carbon nucleus of interest and the molecular plane in angstroms (Å). For the optimized structure of the 2-methylimidazole-benzene complex derived from DFT calculations (see Figure S9), R was determined to be 3.27 Å, which yielded a shielding effect of approximately 1.5 ppm. In the case of CH- π bond, the C-H bond is anticipated to elongate owing to the

attraction of the proton by the π electrons, resulting in an anti-shielding effect on the ^{13}C nucleus. Scheiner reported an anti-shielding effect of 200 ppm/Å for the tetrel bond of a methyl group [48]. To account for the observed downfield shift, an average elongation of approximately 0.01–0.015 Å in the C–H bond is necessary, which is consistent with the expected bond-length increase associated with a strong CH– π bond. In summary, the downfield shifts in the resonance lines resulting from the adsorption of benzene and cyclic alkenes can be attributed to the CH– π bond between the CH protons of the methyl group and imidazole ring and π electrons of the adsorbed molecules.

3.4 Local structure of molecular passage through 6MR

This study elucidates the intermolecular interactions that dictate molecular orientation during the adsorption of bulky molecules onto ZIF-8, particularly as these molecules navigate through 6MR apertures. The results suggest a specific configuration for benzene and six-membered-ring alicyclic hydrocarbons as they traverse the 6MR aperture (Fig. 9). The CH– π bond are pivotal in influencing the molecular orientation of benzene and cyclic alkenes during their passage through the aperture. As these molecules approach the CH bottleneck of the 6MR, they form CH– π bonds with the CH groups of the 2-methylimidazole ring at this bottleneck. This highly directional CH– π bonding results in a nearly perpendicular (T-shaped) alignment of the imidazole ring and adsorbed molecule. This arrangement ensured that the adsorbed molecules maintained a configuration that minimized their molecular cross-sectional areas (molecular planes) upon entering the 6MR aperture. After successfully passing through the aperture, the adsorbed molecule is stabilized by forming an additional CH– π bond with the methyl group on the methyl bottleneck side (Figure S11).

In contrast, cyclic alkanes such as cyclohexane primarily engage in dispersion-force interactions. The optimal molecular orientation for enhancing these interactions occurs when the plane of the adsorbed molecule is parallel to the 2-methylimidazole ring, as this alignment maximizes the contact area between them. Consequently, a molecule approaching the CH bottleneck side of 6MR is captured by dispersion force interactions with the methyl group and quaternary carbon in the 2-methylimidazolate moieties. This results in the adsorbed molecule orienting its minimum molecular cross-section (molecular face) towards the 6MR aperture, thereby facilitating its entry. After passing through the 6MR aperture, the molecule stabilizes through dispersive force interactions with the three methyl groups on the methyl bottleneck side. Cyclohexene, being a cyclic alkene, also exhibits significant contributions from dispersion force

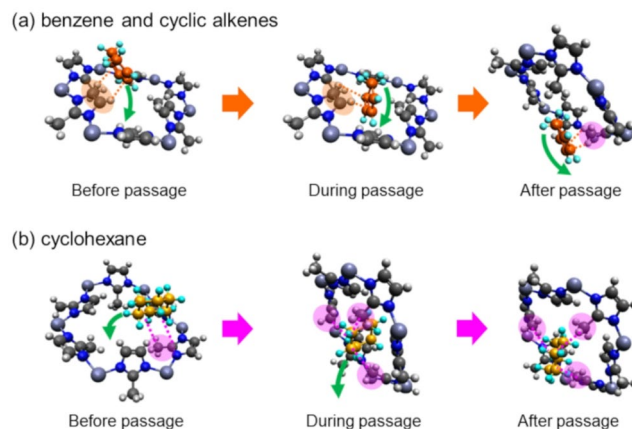


Fig. 9 Model of the configuration of benzene and six-membered-ring alicyclic hydrocarbons traversing the 6MR aperture in ZIF-8

interactions due to its substantial proportion of saturated hydrocarbon components, in addition to the CH– π interactions (see Figure S10).

Finally, ZIF-8 gave somewhat different uptake rate for 1,3-CHD and 1,4-CHD. This result is of particular interest because it highlights the possibility of ZIF-8 as a molecular sieve for isomer separation. Further investigation is necessary to elucidate these detailed mechanisms. However, the different uptake rate between 1,3-CHD and 1,4-CHD may involve the symmetry of the phonon modes associated with the swing effect of the bridging ligands that constitute the 6MR, as well as the symmetry of the adsorbate molecule. Additionally, considering the aspect of configuration diffusion, these molecules might be the different internal arrangement of π electrons through the CH– π bond, which controls the molecular orientation with respect to the 6MR aperture.

4 Conclusion

In this study, we investigated the adsorption kinetics of benzene and six-membered-ring alicyclic hydrocarbons on ZIF-8, to clarify the interactions between the adsorbed molecules and linkers in relation to the adsorption behavior of larger molecules. Monte Carlo simulations were employed to elucidate the local structural dynamics of the molecules as they navigated through the six-membered ring aperture of ZIF-8. Additionally, the FT-IR and ^{13}C cross-polarization magic angle spinning nuclear magnetic resonance (CP/MAS NMR) were used to identify the adsorbate-adsorbate interactions that influence molecular orientation during transit through the aperture.

The findings indicate that the diffusion of benzene and other six-membered ring hydrocarbons through the six-membered ring aperture is predominantly governed by a significant negative activation entropy ranging from –170 to

$-220 \text{ J K}^{-1} \text{ mol}^{-1}$. This observation suggests a pronounced effect of the molecular orientation on the transit of adsorbate molecules through the apertures. Monte Carlo simulations further elucidated this phenomenon, revealing that the aperture diameter significantly affected the activation energy required for the molecular passage. Order parameter analysis revealed that the molecular orientation was tightly constrained and aligned in a manner that minimizes the energy barrier for traversing the aperture. Spectroscopic analyses employing FT-IR and ^{13}C CP/MAS NMR techniques identified a specific CH- π interaction between the π electrons of the adsorbed molecules and the methyl group and imidazole ring CH of the 2-methylimidazolate linker. For unsaturated cyclic hydrocarbons possessing π electrons, the CH- π interaction predominantly governs the molecular orientation during aperture transit. Conversely, saturated cyclic hydrocarbons are primarily influenced by dispersion force interactions with the methyl groups of the bridging ligand. These orientation-controlling interactions are anticipated to play a critical role in governing molecular adsorption and diffusion behaviors of other metal-organic frameworks (MOFs), porous coordination polymers (PCPs), and porous organic materials (POMs) composed of organic materials.

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Data availability The data presented in this article will be available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

Ethical approval Not applicable.

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